# Motor Speed Chart

Synchronous speed for each pole count at 60 Hz and 50 Hz, and how much torque a horsepower is worth at that speed.

**An AC motor's synchronous speed is 120 times the frequency divided by the number of poles.** At 60 Hz that is 3,600 rpm for two poles and 1,800 for four. An induction motor runs a few percent below these figures at full load; the shortfall is its slip.

## Synchronous speed and torque

| Poles | 60 Hz (rpm) | 50 Hz (rpm) | lb-ft per hp at 60 Hz | N·m per kW at 50 Hz |
|---|---|---|---|---|
| **2** | 3,600 | 3,000 | 1.46 | 3.18 |
| **4** | 1,800 | 1,500 | 2.92 | 6.37 |
| **6** | 1,200 | 1,000 | 4.38 | 9.55 |
| **8** | 900 | 750 | 5.84 | 12.73 |
| **10** | 720 | 600 | 7.29 | 15.92 |
| **12** | 600 | 500 | 8.75 | 19.10 |

Torque per horsepower is at the 60 Hz synchronous speed; torque per kW is at the 50 Hz speed. Actual full-load torque is slightly higher, because the motor runs slightly slower.

## How the values are calculated

$$N = 120 \times f / \text{poles}$$

$$T\ (\text{lb-ft}) = 5{,}252 \times \text{hp} / \text{rpm}$$

$$T\ (\text{N·m}) = 9{,}549 \times \text{kW} / \text{rpm}$$

Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

## What this chart does not tell you

These are synchronous speeds. The speed an induction motor actually runs at depends on its load and its design, and comes from the nameplate. Torque figures are running torque at rated power; starting and breakdown torque are separate values set by the motor's design letter.

**This is a reference, not a design.** It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.